General Mathematics — Exercise Sheet 15

Complex arithmetic, roots and rotations

Variant: Statements

Exercise 1 (Complex equations and an excluded denominator). Solve both tasks, retaining all solutions.

1. Find every $z \in \mathbb{C} \setminus \{\mathrm{i}\}$ satisfying $(z+1)/(z-\mathrm{i}) = 2-\mathrm{i}$.
2. Find every $z \in \mathbb{C}$ satisfying $z + \overline{z} = 4$ and $z\overline{z} = 13$.

Exercise 2 (Six roots and their positions). Find every solution of $w^6 + 27 = 0$. List the roots in Cartesian form and classify which have positive real part, which lie on the imaginary axis and which lie on the real axis.

Exercise 3 (A finite sum of squared cosines). Use the seventh roots of unity and the school double-angle identity to evaluate exactly

$$S = \sum_{k=1}^{6} \cos^2\left(\frac{\pi k}{7}\right).$$

Do not replace the individual terms by decimal approximations.

Exercise 4 (Using the real matrix representation). Use the local two-by-two matrix operations and the map Φ of Lecture 15. Consider

$$A = \begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Decide whether each is $\Phi(z)$ for some $z \in \mathbb{C}$. Use complex multiplication to compute A^4. Find the pair (x, y) sent by A to $(7, 1)$, using complex division, and verify both real equations.

Exercise 5 (Composing rotations about different centres). A rotation about $c \in \mathbb{C}$ through angle φ has map $z \mapsto c + \operatorname{cis}\varphi\,(z-c)$. Consider the quarter-turn T about $1+\mathrm{i}$ and the half-turn S about 1, with maps

$$T(z) = \mathrm{i}(z - (1+\mathrm{i})) + (1+\mathrm{i}), \qquad S(z) = -(z-1) + 1.$$

Find $T \circ S$ and $S \circ T$. Express each as a rotation by giving its centre and an angle in $[0, 2\pi)$. Decide whether the two compositions agree.